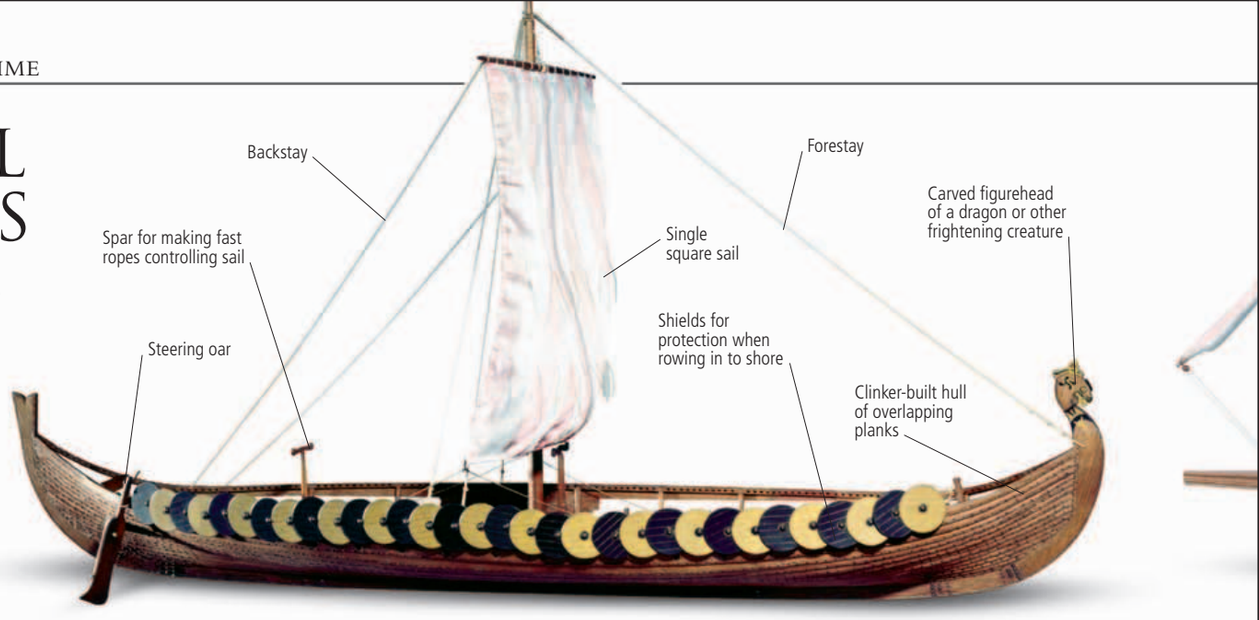
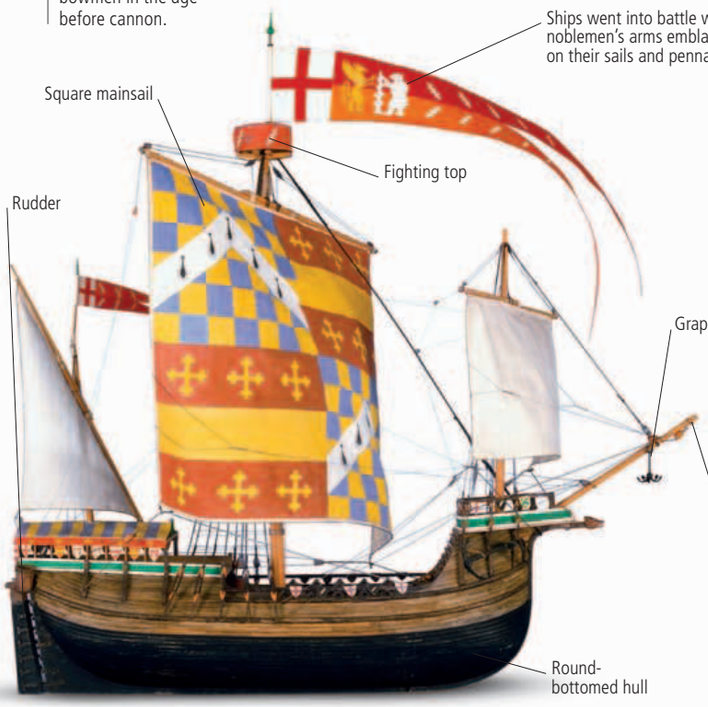
MEDIEVAL WARSHIPS

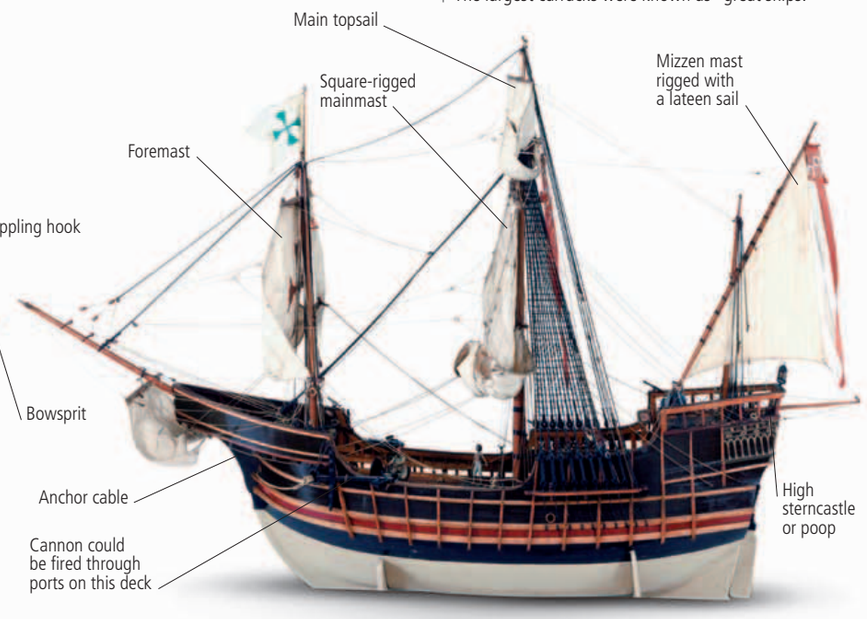
VIKING LONGSHIP (c.1000) Propelled by oars or sail, the longship was capable of ocean voyages and of sailing up rivers. The Vikings rarely fought sea battles as such, but their ships played a key part in raids and invasions. Their shallow draft made for easy beaching and launching.



COG (c.1485) Essentially a merchant ship, the cog was used for battles at sea in northern Europe throughout the later Middle Ages. Its high castles fore and aft made it an excellent platform for bowmen in the age before cannon.



CARRACK (c.1490) The carrack was a three- or four-masted ship used for long-distance trade and warfare. This is Christopher Columbus's flagship *Santa Maria*. More stable, but less maneuverable than the caravel, the carrack was the forerunner of the galleon. The largest carracks were known as "great ships."



THE DESIGN of medieval warships developed along very different lines in the Mediterranean and the seas of northern Europe. Mediterranean states continued to use oared galleys for warfare. Their shipbuilders made hulls that were carvel-built (had flush-fitting planks) and, under the influence of Arab sailors from the Indian Ocean, adopted the triangular lateen sail instead of the square sail. In northern Europe the Viking longship, the prime warship of the early Middle Ages, was clinker-built (had overlapping planks that needed less caulking than carvel-built hulls) and had a square sail. The longship had oars, like a Mediterranean war galley, but was better suited to rough seas.

North European shipbuilders also used a clinker-built hull and a square sail on the sturdy round cargo ship known as a "cog." Cogs came to be used in warfare, in the process growing "castles" at both ends of the ship, raised platforms from which soldiers and archers could rain missiles down upon enemy ships and where they could resist attempts to board. Although oared galleys continued to play a part in warfare in northern Europe, sailing ships took precedence.



Shipborne warriors
This 8th-century carved stone from the Swedish island of Gotland shows Viking warriors aboard a longship sailing to Valhalla after a life of raids and invasions.

HYBRID SHIPS

In the 15th century a fruitful cross-fertilization between the Mediterranean and northern European traditions took place. The three- or four-masted carrack combined lateen and square sails to create a larger, more maneuverable descendant of the cog. At the same time the smaller two- or three-masted caravel evolved, mostly with lateen sails but sometimes with a square sail as well. By 1500 the carrack was the dominant warship of northern Europe. It provided a stable platform for cannon, used in large numbers on ships from the late 15th century. With gun ports cut into its hull, the carrack grew into the "great ship"—for example, *Henry Grâce à Dieu*, launched in 1514, which carried some 180 cannon, distributed along gun decks and in prodigiously high castles.